

**САНКТ-ПЕТЕРБУРГСКИЙ НАЦИОНАЛЬНЫЙ
ИССЛЕДОВАТЕЛЬСКИЙ УНИВЕРСИТЕТ ИТМО**

Отчет по лабораторной работе №1

по дисциплине: “Администрирование компьютерных сетей”

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Санкт-Петербург

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Задача

Подключить к тестовому сервису Nextcloud мониторинг + логирование.

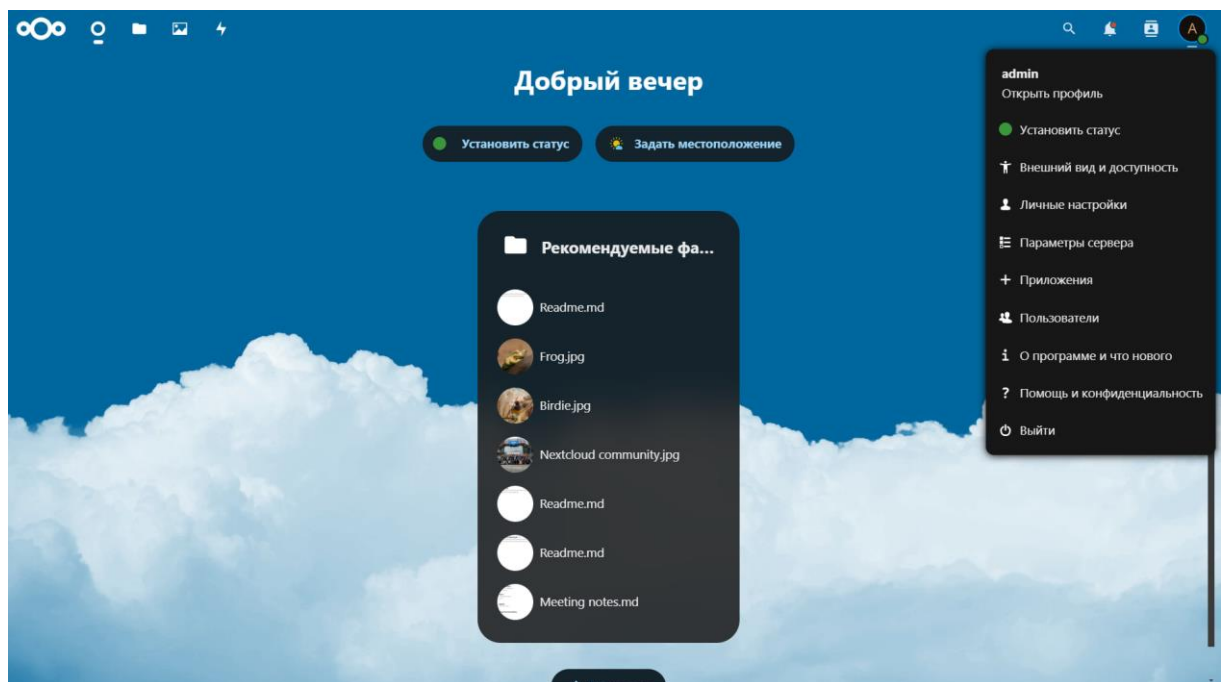
Осуществить визуализацию через Grafana

1. Логирование

Я создал файлы `docker-compose.yml` и `promtail_config.yml` и запустил их командой `docker-compose up -d`

```
kazar@LAPTOP-4M4USRCD MINGW64 ~/Desktop/Admin Network (dev)
$ docker-compose up -d
[+] Running 76/7
  ✓ promtail Pulled                                254.1s
  ✓ postgres-zabbix Pulled                          416.4s
  ✓ zabbix-web-nginx-pgsql Pulled                    513.1s
  ✓ grafana Pulled                                  442.1s
  ✓ nextcloud Pulled                                722.7s
  ✓ zabbix-server Pulled                            313.7s
  ✓ loki Pulled                                      208.7s
[+] Running 10/10
  ✓ Network adminnetwork default Created            0.0s
  ✓ Volume "adminnetwork_nc-data" Created            0.0s
  ✓ Volume "adminnetwork_zabbix-db" Created          0.0s
  ✓ Container postgres-zabbix Started                1.1s
  ✓ Container nextcloud Started                      1.8s
  ✓ Container grafana Started                        1.7s
  ✓ Container loki Started                           1.4s
  ✓ Container promtail Started                       1.3s
  ✓ Container zabbix-back Started                    1.7s
  ✓ Container zabbix-front Started                   1.6s
```

Зашел на веб интерфейс через внешний порт, указанный в compose файле (`localhost:8080`), создал учетную запись



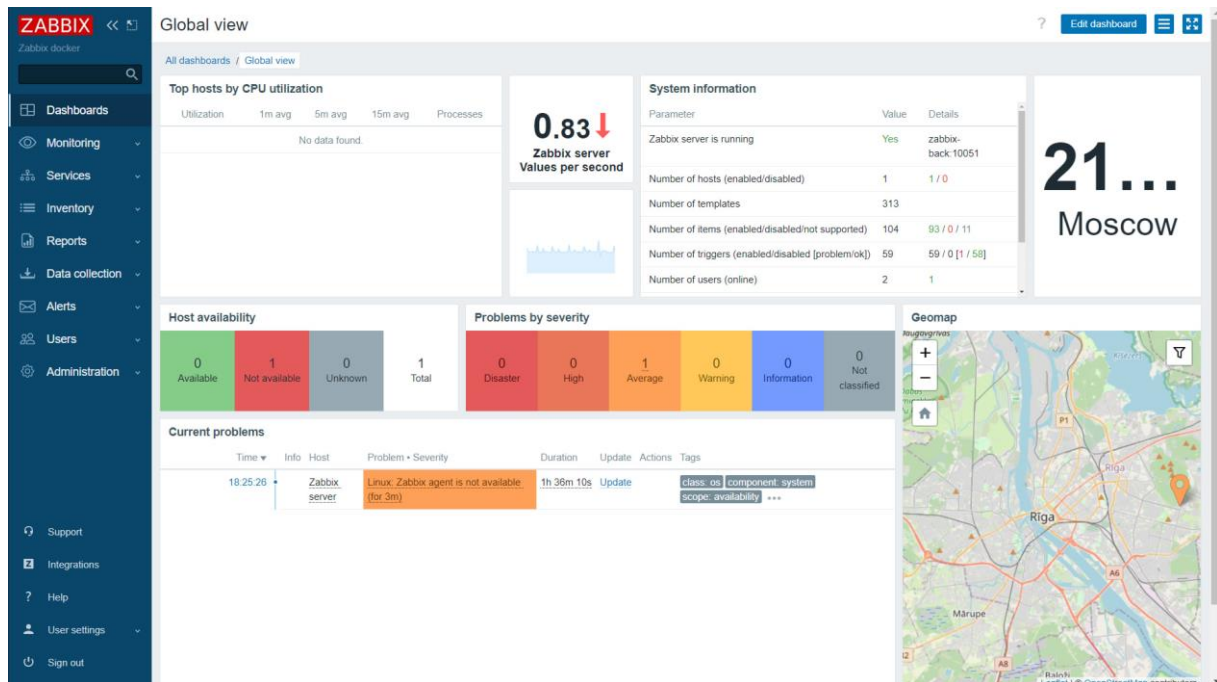
После инициализации Nextcloud проверил в логах promtail, что он “подцепил” нужный нам файл:

```
kazar@LAPTOP-4M4USRCD MINK664 ~/Desktop/Admin Network (dev)
$ docker logs 902954a8db829f9bdfef5080eef37bad353cf80eb228c881b8b487536516633a
level=info ts=2025-01-09T16:23:58.328195052Z caller=promtail.go:133 msg="Reloading configuration file" md5sum=2c8ba9ad5647669e9f64bdad5fed7eae
level=info ts=2025-01-09T16:23:58.329832393Z caller=server.go:322 http[::]:9080 grpc[::]:43447 msg="server listening on addresses"
level=info ts=2025-01-09T16:23:58.330175121Z caller=main.go:174 msg="Starting Promtail" version="(version=2.9.0, branch=HEAD, revision=2feb64f69)"
level=warn ts=2025-01-09T16:23:58.330270256Z caller=promtail.go:263 msg="enable watchConfig"
level=info ts=2025-01-09T16:24:03.330505794Z caller=filetargetmanager.go:361 msg="Adding target" key="/opt/nc_data/*.log:{job=\"nextcloud_logs\"}"
level=info ts=2025-01-09T17:44:23.096768983Z caller=filetarget.go:313 msg="watching new directory" directory=/opt/nc_data
level=info ts=2025-01-09T17:44:23.099258992Z caller=tailer.go:145 component=tailer msg="tail routine: started" path=/opt/nc_data/nextcloud.log
ts=2025-01-09T17:44:23.099573231Z caller=log.go:168 level=info msg="Seeked /opt/nc_data/nextcloud.log - &{Offset:0 Whence:0}"
level=info ts=2025-01-09T17:45:07.603020872Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/htaccessstest.txt op=CREATE
level=info ts=2025-01-09T17:45:07.613613256Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/owncloud.db op=CREATE
level=info ts=2025-01-09T17:45:07.620894974Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/owncloud.db-journal op=CREATE
level=info ts=2025-01-09T17:45:07.632153681Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/owncloud.db-wal op=CREATE
level=info ts=2025-01-09T17:45:07.632228661Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/owncloud.db-shm op=CREATE
level=info ts=2025-01-09T17:45:10.487121161Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/ocdata op=CREATE
level=info ts=2025-01-09T17:45:10.86960449Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/admin op=CREATE
level=info ts=2025-01-09T17:45:25.827439034Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/owncloud.db-wal op=CREATE
level=info ts=2025-01-09T17:45:25.827487582Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/owncloud.db-shm op=CREATE
level=info ts=2025-01-09T17:45:25.910830947Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/appdata/oci31pfci3rs op=CREATE
level=info ts=2025-01-09T17:45:26.079190155Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/owncloud.db-wal op=CREATE
level=info ts=2025-01-09T17:45:26.079258395Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/owncloud.db-shm op=CREATE
level=info ts=2025-01-09T17:45:26.387372169Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/owncloud.db-wal op=CREATE
level=info ts=2025-01-09T17:45:26.387432431Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/owncloud.db-shm op=CREATE
level=info ts=2025-01-09T17:45:26.725251677Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/owncloud.db-wal op=CREATE
level=info ts=2025-01-09T17:45:26.725302172Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/owncloud.db-shm op=CREATE
level=info ts=2025-01-09T17:45:44.129430089Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/owncloud.db-wal op=CREATE
level=info ts=2025-01-09T17:45:44.129525167Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/owncloud.db-shm op=CREATE
level=info ts=2025-01-09T17:45:44.328052647Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/owncloud.db-wal op=CREATE
level=info ts=2025-01-09T17:45:44.328155543Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/owncloud.db-shm op=CREATE
level=info ts=2025-01-09T17:45:44.702272088Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/owncloud.db-wal op=CREATE
```

2. Мониторинг

Настроил Zabbix, подключился к веб-интерфейсу по `http://localhost:8082`

Сделал Import кастомного шаблона (темплейта) для мониторинга nextcloud.



Зашел на контейнер под юзером `www-data` и выполнили команду `php occ config:system:set trusted_domains 1 --value="nextcloud"`

```
kazar@LAPTOP-8M4USRCD MINGW64 ~/Desktop/Admin Network (dev)
$ docker exec -u www-data -it nextcloud bash
www-data@70bc211122cf:~/html$ php occ config:system:set trusted_domains 1 --value="nextcloud"
System config value trusted_domains -> 1 set to string nextcloud
www-data@70bc211122cf:~/html$
```

Добавляем хост и начинаем мониторинг:

Hosts ? [Create host](#) [Import](#)

✓ Host added

Filter

Host groups

type here to search

Select

Templates

type here to search

Select

Name

DNS

IP

Port

Status

Any

Enabled

Disabled

Monitored by

Any

Server

Proxy

Proxy

Select

Tags

And/Or

Or

tag

Contains

value

Remove

Add

Apply

Reset

☐	Name	Items	Triggers	Graphs	Discovery	Web	Interface	Proxy	Templates	Status	Availability	Agent encryption	Info	Tags
☐	Nextcloud Server	Items 1	Triggers 1	Graphs	Discovery	Web			Test ping template	Enabled		None		
☐	Zabbix server	Items 104	Triggers 59	Graphs 19	Discovery 5	Web	127.0.0.1:10050		Linux by Zabbix agent, Zabbix server health	Enabled	ZBX	None		

0 selected [Enable](#) [Disable](#) [Export](#) [Mass update](#) [Delete](#) Displaying 2 of 2 found

ZABBIX Zabbix docker

Dashboards

Monitoring

Problems

Hosts

Latest data

Maps

Discovery

Services

Inventory

Reports

Data collection

Alerts

Users

Administration

Support

Integrations

Help

User settings

Sign out

Latest data

Filter

Host groups

type here to search

Select

Hosts

type here to search

Select

Name

Tags

And/Or

Or

tag

Contains

value

Remove

Add

Show tags

None

1

2

3

Tag name

Full

Shortened

None

Tag display priority

comma-separated list

Show details

Save as

Apply

Reset

Subfilter affects only filtered data

HOSTS

Nextcloud Server

Zabbix server +104

TAGS

TAG VALUES

DATA

With data

Without data

☐	Host	Name	Last check	Last value	Change	Tags	Info
☐	Nextcloud Server	Nextcloud_ping.service	48s	healthy			History

0 selected [Display stacked graph](#) [Display graph](#) [Execute now](#) Displaying 1 of 1 found

Zabbix 6.4.20. © 2001–2024, Zabbix SIA

По желанию включил в некстклауде *maintenance mode* (php oss maintenance:mode --on в контейнере), проверил что сработал триггер, выключил режим обратно (php oss maintenance:mode --off), убедился, что проблема помечена как “решенная”.

The screenshot shows the Zabbix web interface, specifically the 'Problems' section. The left sidebar contains navigation links for Dashboards, Monitoring (Problems, Hosts, Latest data, Maps, Discovery), Services, Inventory, Reports, Data collection, Alerts, Users, Administration, Support, Integrations, Help, User settings, and Sign out. The main content area is titled 'Problems' and includes filters for Host groups, Hosts, Triggers, and Problem. It also has checkboxes for Severity (Not classified, Warning, High, Information, Average, Disaster), Age less than (14 days), Show symptoms, Show suppressed problems, and Show unacknowledged only. On the right, there are filters for Host inventory, Tags, Show tags, Tag display priority, Show operational data, Compact view, Show details, and Highlight whole row. Below these filters is a table of problems:

Time	Severity	Recovery time	Status	Info	Host	Problem	Duration	Update	Actions	Tags
20:41:31	Disaster				Nextcloud Server	Nextcloud is in maintenance mode	43s	Update		
20:00										
18:25:26	Average		PROBLEM		Zabbix server	Linux: Zabbix agent is not available (for 3m)	2h 16m 48s	Update		class: os component: system scope: availability ***

At the bottom, it says '0 selected' and 'Mass update'. The footer indicates 'Zabbix 6.4.20. © 2001–2024, Zabbix SIA'.

ZABBIX

Zabbix docker

Monitoring

Problems

Hosts

Latest data

Maps

Discovery

Services

Inventory

Reports

Data collection

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Users

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Support

Integrations

Help

User settings

Sign out

Problems

Recent problems

Problems

History

Host groups

type here to search

Select

Hosts

type here to search

Select

Triggers

type here to search

Select

Problem

Severity

☐ Not classified

☐ Warning

☐ High

☐ Information

☐ Average

☐ Disaster

Age less than

☐ 14

days

Show symptoms

☐

Show suppressed problems

☐

Show unacknowledged only

☐

Host inventory

Type

Remove

Tags

And/Or

Or

tag

Contains

value

Remove

Show tags

None

1

2

3

Tag name

Full

Shortened

None

Tag display priority

comma-separated list

Show operational data

None

Separately

With problem name

Compact view

☐

Show details

☐

Show timeline

☒

Highlight whole row

☐

Save as

Apply

Reset

☐	Time	Severity	Recovery time	Status	Info	Host	Problem	Duration	Update	Actions	Tags
☐	20:41:31	Disaster	20:43:31	RESOLVED		Nextcloud Server	Nextcloud is in maintenance mode	2m	Update		
☐	18:25:26	Average		PROBLEM		Zabbix server	Linux: Zabbix agent is not available (for 3m)	2h 18m 18s	Update		class: os component: system scope: availability ...

0 selected

Mass update

Displaying 2 of 2 found

Zabbix 6.4.20. © 2001–2024, Zabbix SIA

3. Визуализация

В терминале выполнил команду `docker exec -it grafana bash -c "grafana cli plugins install alexanderzobnin-zabbix-app"`, затем `docker restart grafana`

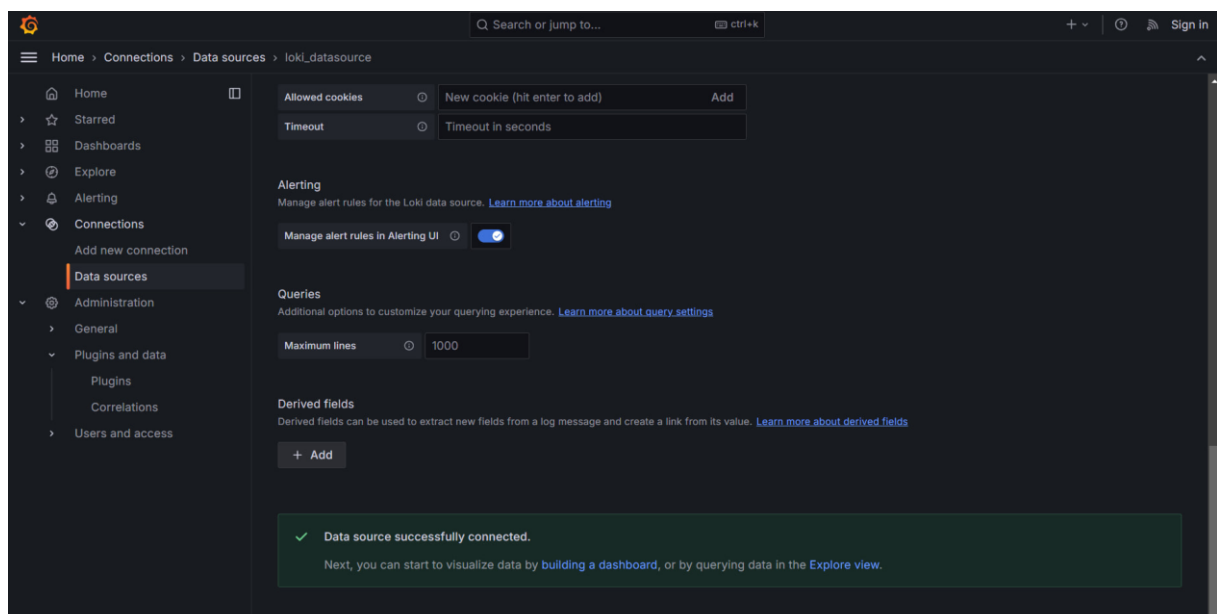
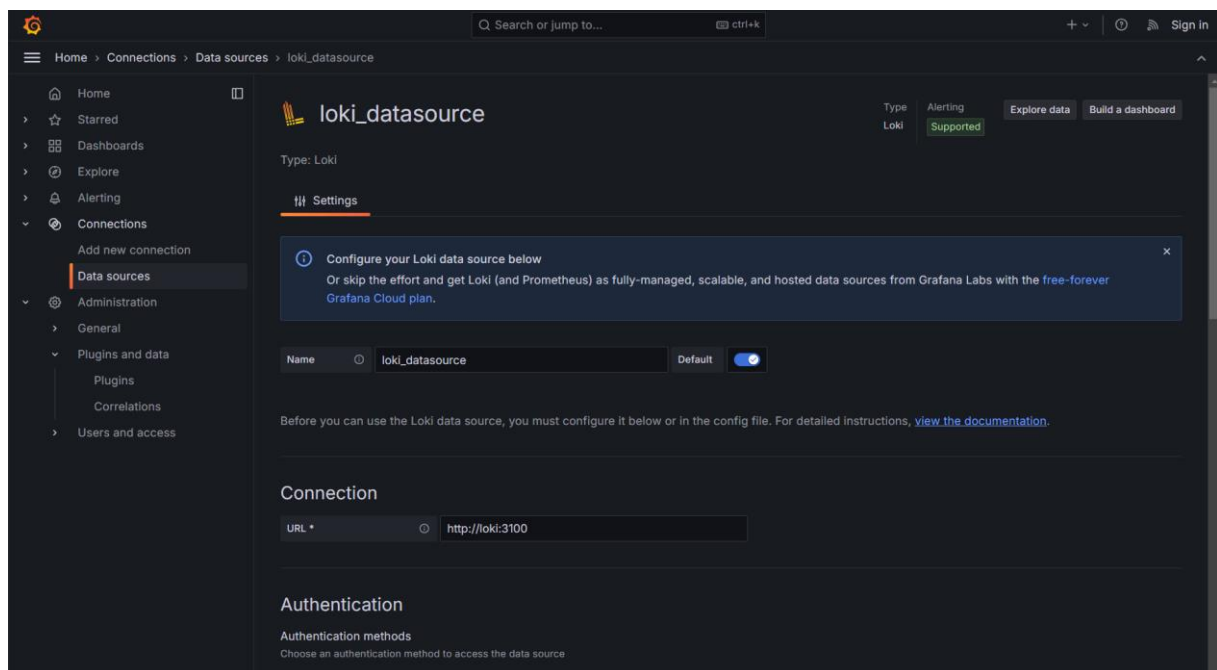
```
kazar@LAPTOP-4M4USRCD MINGW64 ~/Desktop/Admin Network (dev)
• $ docker exec -it grafana bash -c "grafana cli plugins install alexanderzobnin-zabbix-app"
✓ Downloaded and extracted alexanderzobnin-zabbix-app v4.6.1 zip successfully to /var/lib/grafana/plugins/alexanderzobnin-zabbix-app

Please restart Grafana after installing or removing plugins. Refer to Grafana documentation for instructions if necessary.

kazar@LAPTOP-4M4USRCD MINGW64 ~/Desktop/Admin Network (dev)
• $ docker restart grafana
grafana

kazar@LAPTOP-4M4USRCD MINGW64 ~/Desktop/Admin Network (dev)
```

Зашли в графану (по `http://localhost:3000/`), нашел там Zabbix и активировал. Подключили Loki к Grafana, сохранил и проверил подключение.

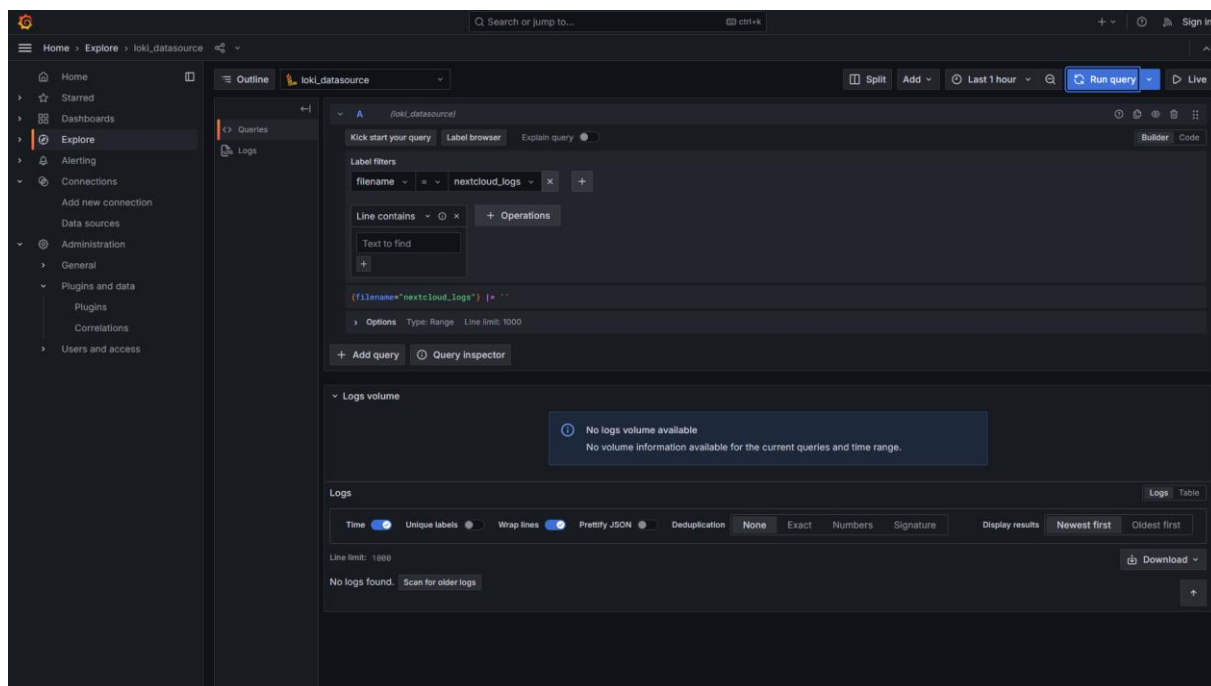


Так же подключил Zabbix

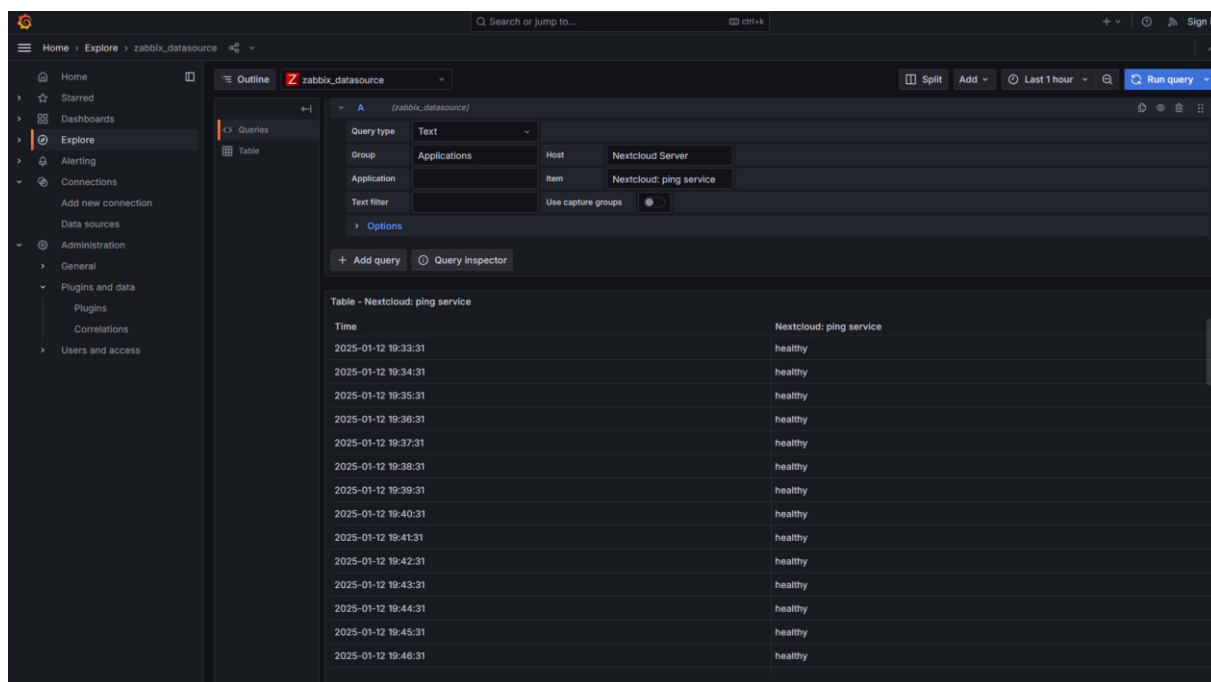
The screenshot shows the Grafana interface for configuring a Zabbix data source. The left sidebar contains navigation links: Home, Starred, Dashboards, Explore, Alerting, Connections, Data sources (selected), Administration, General, Plugins and data, Plugins, Correlations, and Users and access. The main panel is titled 'zabbix_datasource' and shows the 'Settings' tab. It includes a 'Name' field set to 'zabbix_datasource', a 'Type' dropdown set to 'Zabbix', and an 'Alerting' status set to 'Supported'. Below this, there is a 'Connection' section with a 'URL' field set to 'http://zabbix-front:8080/api_jsonrpc.php'. An 'Authentication' section shows a dropdown menu set to 'No Authentication'. At the bottom, there are 'TLS settings' with a checkbox for 'Add self-signed certificate'.

This screenshot shows the 'Advanced HTTP settings' for the Zabbix data source. It includes fields for 'Allowed cookies' (with a 'New cookie (hit enter to add)' button and an 'Add' button), 'Timeout' (set to 'Timeout in seconds'), 'Cache TTL' (set to '1h'), and 'Timeout' (set to '30'). There is a 'Trends' section with a toggle for 'Enable Trends' (checked), an 'After' field set to '7d', and a 'Range' field set to '4d'. Below this is a 'Direct DB Connection' section with a toggle for 'Enable Direct DB Connection' (checked). At the bottom, there are two checkboxes: 'Disable acknowledges for read-only users' and 'Disable data alignment', both of which are checked. A green notification box at the bottom states 'Zabbix API version: 6.4.20' and provides instructions on how to visualize data. At the very bottom, there are 'Delete' and 'Save & test' buttons.

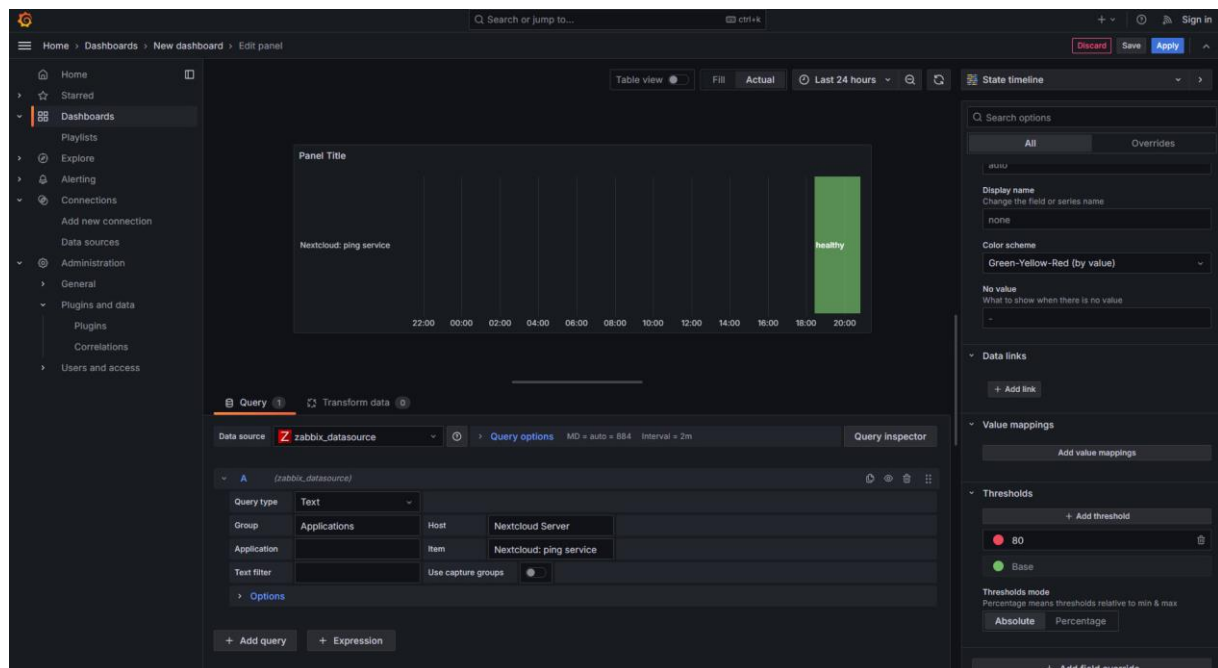
Добавил в графану data source Loki (логов почему то нет, причины не нашел)



Добавил в графану data source Zabbix



Создал дашборд в Grafana с использованием датасурсов Zabbix и Loki



Из-за отсутствия логов не могу сделать больше, иначе хотя бы повторил бы тот, что у вас в примере

4. Вопросы

Чем SLO отличается от SLA?

SLA — это договор с клиентом с обязательствами и штрафами.

SLO — это внутренняя цель сервиса для контроля выполнения SLA.

Чем отличается инкрементальный бэкап от дифференциального?

Инкрементальный — сохраняет изменения с последнего бэкапа (быстрее, меньше данных).

Дифференциальный — сохраняет изменения с последнего полного бэкапа (больше данных, но проще восстановить).

В чем разница между мониторингом и observability?

Мониторинг — это отслеживание заранее заданных метрик.

Observability — это более глубокое понимание системы с анализом метрик и логов.